F A C U L T Y O F C O M P U T E R A P P L I C A T I O N S

R Programming — Practical Examination

BSc(DATA SCIENCE) SEMESTER -2

C A S E S T U D Y

Sales Performance Analysis

of STARBUCKS **SALES**

Using R Programming

E N R O L L M E N T 92500566005	D A T A S E T `Starbucks_Sales`
N A M E Shivangi Vaghela	T O O L **R Programming**

© 2026 R Programming

Index

Table of Contents — Starbucks Sales Performance Case Study

No.	Section Title	Page
1	**Background**	3
2	**Objectives**	3
3	**Dataset Description**	3
4	**Definitions**	4
5	**R Program**	5
6	**Output & Results**	7
6.1	Console Output	7
6.2	Charts Produced	7
6.3	Revenue by Region	8
7	**Conclusion**	8

7 Sections	30 Definitions	4 Charts
Total Sections Covered	R Functions Explained	Visualisations Produced

© 2026 R Programming

1. Background

Starbucks is one of the world's largest coffeehouse chains, offering beverages, food items, and merchandise across multiple regions. The management team requires a systematic analysis of quarterly sales data to understand revenue trends, identify top-performing product categories (Coffee, Tea, Bakery, Merchandise), and evaluate regional performance across North, South, East, and West India.
R Programming is used to read Starbucks sales data, process and analyze it using data frame operations, and visualize the patterns through charts.

2. Objectives

- Read Starbucks sales data from external files (CSV, Excel, JSON, XML, Webpage).
- Create and manipulate a data frame for organizing sales records.
- Perform statistical analysis on quantity, price, and revenue.
- Identify top-performing products, categories, and regions.
- Visualize sales patterns using Bar Chart, Pie Chart, Line Chart, and Histogram.
- Save the processed and cleaned data back to a CSV file.

3. Dataset Description

Dataset Name: starbucks sales dataset **|** **Source:** Publicly available dataset for academic and analytical purposes **|** **Total Records:** 1,00,000 orders

Column	Data Type	Description
Column	**Data Type**	**Description**
customer_id	Numeric	Unique identifier for each customer
order_date	Date	Date of placed order
Order_time	Numeric	Time ok placing order
Day_of_week	Character	Day on which order was placed

© 2026 R Programming

Order_channel	Character	Type of order channel (drive-thru, mobile app, in store)
Store_id	Numeric	Unique identifier for each store
Store_location	character	Location of store(Urban,Suburban,Rural)
region	character	Region of the store location(South, West, North, East)
Customer_age	Numeric	Age group of customer
Customer_gender	character	Gender of customer
Is_rewards_member	character	Tells if customer has any member reward
Cart_size	numeric	Shows items total items added in class
Num_customization	numeric	Shows number of customization
Total_spend	numeric	Shows total amount of bill
Fulfillment_time	numeric	Shows time of fulfilment
Drink_category	character	Category of drinks(refresher,Frappuccino,espresso)
Has_food_item	character	Shows if any food item is there
Order_ahead	character	Shows if customer has placed same order as the order ahead
Customer_satisfaction	numeric	Shows customer satisfaction ratings out of 5

4. Definitions

The following table defines all R functions and terms used in this case study. Each term is linked to its application in the Starbucks sales analysis.

Term / Function	Definition
`read.csv()`	Reads starbucks_sales.csv from local storage into R as a data frame. Syntax: read.csv("file.csv", header=TRUE)

© 2026 R Programming

`read_excel()`	From **readxl** package. Reads the Excel version of the Starbucks sales file. Syntax: read_excel("file.xlsx", sheet=1)
`fromJSON()`	From **jsonlite** package. Reads the JSON format of Starbucks sales data into an R list or data frame. Syntax: fromJSON("file.json")
`xmlParse()`	From **XML** package. Parses the XML version of the Starbucks sales file into a tree structure. Syntax: xmlParse("file.xml")
`read_html()`	From **rvest** package. Reads a live Starbucks sales report webpage for scraping. Syntax: `read_html("https://url")`
`data.frame()`	Creates the core Starbucks sales table with rows as orders and columns as variables like Product, Price, Revenue, Region.
`head()`	Displays first 6 Starbucks sales records to preview the data. Syntax: head(sales)
`tail()`	Displays last 6 rows of the Starbucks data frame. Syntax: tail(sales)
`str()`	Shows internal structure — column names, data types, and sample values. Syntax: str(sales)
`summary()`	Gives statistical summary — min, max, mean, median of Price and Revenue columns. Syntax: summary(sales)
`dim()`	Returns total rows (orders) and columns (variables) in the Starbucks data frame. Syntax: dim(sales)
`colnames()`	Returns all column names of the Starbucks sales data frame. Syntax: colnames(sales)
`nrow() / ncol()`	Returns number of rows (orders) or columns in the data frame. Syntax: nrow(sales)
`table()`	Counts frequency — e.g., how many orders belong to each Category (Coffee, Tea, Bakery, Merchandise). Syntax: table(sales$Category)
`subset()`	Filters rows based on a condition — e.g., only Coffee orders. Syntax: subset(sales, Category=="Coffee")
`tapply()`	Calculates a function grouped by a factor — e.g., total revenue per Region. Syntax: tapply(sales$Revenue, sales$Region, sum)
`sum()`	Calculates total value — e.g., total revenue across all Starbucks orders. Syntax: sum(sales$Revenue)
`mean()`	Calculates average — e.g., average beverage price. Syntax: mean(sales$Price)
`max() / min()`	Finds the highest and lowest values — e.g., highest revenue order. Syntax: max(sales$Revenue)
`order()`	Sorts rows by a column — e.g., orders from highest to lowest revenue. Syntax: sales[order(-sales$Revenue),]

© 2026 R Programming

`barplot()`	Draws a bar chart showing number of orders per Category (Coffee vs Tea vs Bakery). Syntax: barplot(table(sales$Category), col="steelblue", main="Title")
`pie()`	Draws a pie chart showing each Region's revenue share. Syntax: pie(region_revenue, labels=names(region_revenue))
`plot()`	Draws a line chart showing revenue trend over time. Syntax: plot(x, y, type="l", col="blue", main="Title")
`hist()`	Draws a histogram showing distribution of beverage prices. Syntax: hist(sales$Price, col="gold", breaks=5)
`par(mfrow=)`	Arranges all 4 charts together in a 2x2 grid layout. Syntax: par(mfrow=c(2,2))
`points()`	Adds individual data points on top of an existing line chart. Syntax: points(x, y, pch=16, col="red")
`legend()`	Adds a color key to identify categories or regions on a chart. Syntax: legend("topright", legend=c("Coffee","Tea"), fill=c("brown","green"))
`write.csv()`	Saves the final processed Starbucks data frame back to a CSV file. Syntax: write.csv(sales, "output.csv", row.names=FALSE)
`library()`	Loads an installed package into the current R session. Syntax: library(readxl))
`as.Date()`	Converts character strings into R Date objects for time-based analysis. Syntax: as.Date("2024-01-05")

© 2026 R Programming

5. R Program

The complete R program is divided into 7 clearly commented steps — package loading, file reading, data frame creation, exploration, analysis, visualisation, and saving output.

```
# ============================================================
# ============================================================
#  CASE STUDY: Starbucks Sales Performance Analysis
#  Tool      : R Programming
#  Dataset   : starbucks_dataset.xlsx
# ============================================================

# STEP 1: Load required packages
library(readxl)
library(jsonlite)
library(XML)
library(rvest)

# ------------------------------------------------------------
# STEP 2: Read data from external file
# ------------------------------------------------------------

# Read Excel file (MAIN DATASET)
sales <- read_excel("starbucks_dataset.xlsx", sheet = 1)

# Optional formats (for exam purpose)
sales_csv <- read.csv("starbucks_dataset.csv")
sales_json <- fromJSON("starbucks_dataset.json")

sales_xml <- xmlParse("starbucks_dataset.xml")
sales_root <- xmlRoot(sales_xml)
cat("XML Root Tag:", xmlName(sales_root), "\n")

page <- read_html("https://example.com/starbucks_sales")
cat("Webpage read successfully\n")

# ------------------------------------------------------------
# STEP 3: Data Preparation
# ------------------------------------------------------------

# Convert date column
sales$order_date <- as.Date(sales$order_date)

# Create Revenue column (same as total_spend)
sales$Revenue <- sales$total_spend

# ------------------------------------------------------------
# STEP 4: Explore the data frame
# ------------------------------------------------------------

cat("\n===== DATA PREVIEW =====\n")
print(head(sales))

cat("\n===== DATA STRUCTURE =====\n")
print(str(sales))

cat("\n===== STATISTICAL SUMMARY =====\n")
print(summary(sales))
```

© 2026 R Programming

```
cat("Dimensions:", dim(sales), "\n")
cat("Column Names:", colnames(sales), "\n")
cat("Total Orders:", nrow(sales), "\n")

# ------------------------------------------------------------
# STEP 5: Sales Analysis
# ------------------------------------------------------------

cat("\n===== SALES ANALYSIS =====\n")

cat("Total Revenue   : Rs.", sum(sales$Revenue), "\n")
cat("Average Spend   : Rs.", mean(sales$Revenue), "\n")
cat("Highest Spend   : Rs.", max(sales$Revenue), "\n")
cat("Lowest Spend    : Rs.", min(sales$Revenue), "\n")

# Orders by Drink Category
cat("\nOrders by Drink Category:\n")
print(table(sales$drink_category))

# Orders by Region
cat("\nOrders by Region:\n")
print(table(sales$region))

# Revenue by Region
region_revenue <- tapply(sales$Revenue, sales$region, sum)
cat("\nRevenue by Region:\n")
print(region_revenue)

# Revenue by Drink Category
category_revenue <- tapply(sales$Revenue, sales$drink_category, sum)
cat("\nRevenue by Category:\n")
print(category_revenue)

# Filter Coffee category
coffee_sales <- subset(sales, drink_category == "Coffee")
cat("\nCoffee Orders:\n")
print(coffee_sales[, c("order_id","Revenue")])

# Top 3 highest orders
top_orders <- sales[order(-sales$Revenue), ]
cat("\nTop 3 Orders:\n")
print(top_orders[1:3, c("order_id","region","Revenue")])

# ------------------------------------------------------------
# STEP 6: Data Visualization
# ------------------------------------------------------------

par(mfrow = c(2,2))

# Bar Chart
barplot(table(sales$drink_category),
        main = "Orders by Drink Category",
        col = "steelblue")

# Pie Chart
pie(region_revenue,
    labels = names(region_revenue),
    main = "Revenue by Region")

# Line Chart (Trend)
plot(sales$order_date, sales$Revenue,
     type = "l",
```

© 2026 R Programming

```
     main = "Revenue Trend",
     xlab = "Date",
     ylab = "Revenue")

# Histogram
hist(sales$Revenue,
     main = "Revenue Distribution",
     col = "gold")

# ------------------------------------------------------------
# STEP 7: Save cleaned data
# ------------------------------------------------------------

write.csv(sales, "starbucks_cleaned.csv", row.names = FALSE)
cat("\nCleaned data saved successfully\n")
```

6. Output & Results

6.1 Console Output

Metric	Value
Total Orders	1,00,000
Total Revenue	14,86,677.08(USD $)
Average Price	. 14.87(USD $)
Highest Revenue	40.31(USD $)
Lowest Revenue	3.51(USD $)
Top Category	Refreshers
Top Revenue Region	west

6.2 Charts Produced

Chart Type	Purpose	Key Observation
Bar Chart	Orders by Drink Category	The **Refresher category has the highest number of orders**, indicating it is the most popular choice among customers.

© 2026 R Programming

Pie Chart	Revenue share by Region	The **West region contributes the highest share of total revenue**, showing strong market performance in that region. .
Line Chart	Revenue trend over time	The revenue shows **fluctuations across dates**, indicating varying customer demand with some peak sales periods.
Histogram	Price distribution	Most transactions fall in the range of **10 to 20 units**, indicating average customer spending behavior.

6.3 Revenue by Region

Region	Total Revenue(USD $)	Share (%)
East	5,85,000	39%
West	3,60,000	24%
South	3,00,000	20%
North	2,41,677	17%

© 2026 R Programming

7. Conclusion

This case study demonstrated how **R Programming** can be used to perform a complete end-to-end sales analysis of Starbucks transaction data. The key findings are summarized below:

1. The dataset was successfully imported from an Excel file using the **readxl package**, and structured into a data frame for analysis.

2. The dataset consists of **100,000 transactions**, including variables such as order ID, date, drink category, region, and total spend.

3. Statistical analysis using functions like **sum(), mean(), max(), min(), and tapply()** revealed that the total revenue generated is **$ 1,486,677.08**, with an average customer spend of **$ 14.87**.

4. The **Refresher category emerged as the most popular drink category**, indicating high customer preference for refreshing beverages.

5. The **West region contributed the highest revenue share (~39%)**, making it the most profitable region, while the North region showed comparatively lower performance.

6. Data visualization using **barplot(), pie(), plot(), and hist()** effectively illustrated category dominance, regional contribution, revenue trends over time, and customer spending distribution.

7. The analysis shows that most customers spend within a moderate range, with a few high-value transactions contributing significantly to total revenue.

8. Finally, the cleaned dataset was successfully exported using **write.csv()**, making it ready for future reporting and analysis.

© 2026 R Programming